THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONAE PRE MOCK JOINT EXAMINATION
FORM FOUR- BIOLOGY

TIME: 3 HOURS **May, 2021**

INSTRUCTIONS.

- This paper consist of sections, A, B and C
- Answer all questions from section A and B and two questions from
- Section C also question number 13 is compulsory.

SECTION A (15 MARKS)

1. For each of the items (i) _(x) choose the correct answer from among the given alternatives and write its letter beside the item number in answer sheet provided.

i. which of the following are parasitic eukaryotic organisms?
A. Amoeba and paramecium
B. Euglena and green algae
C. Trypanosome and plasmodium
D. Brown algae and amoeba
E. Plasmodium and amoeba ( )

ii. Which one of the following is group of animal tissues?
A. Muscles, blood nerve and epithelia
B. Eyes muscles skin and blood
C. Cardiac muscles, skin, liver and blood
D. Liver, pancreas, lymph and ear
E. Blood, muscles, skin and kidney ( )

iii. A Flower which possesses both stamen and carpels is said to be.....
A. Unisexual B. Hermaphrodite C. Monoecism D. Zygomorphic E. Polymorphic ( )

iv. Useful substances are retained in the kidney by....
A. Filtration B. Osmosis C. Selective reabsorption D. Osmo-regulation E. Diffusion ( )

v. A biologists discovered a new cell in a culture. The new cell had a distinct cell wall but it did not have a definite Nucleus. The cell is most likely to be....
A. Fungi B. Prokaryote C. Protozoa D. Virus E. Eukaryote ( )

vi. Oxygen dept occur when.....
A. Too much carbon dioxide is present in the body
B. The rate of respiration exceeds that of photosynthesis in green plants
C. Alcohol is formed in tissues
D. There is insufficient oxygen in the muscles during strenuous exercise
E. There is insufficient lactic acid in muscle tissues ( )

vii. A human ovary produces ova and the following hormones

A. Oestrogen and testosterone
B. progesterone and testosterone
C. Follicle stimulating hormone (F.S.H) and progesterone
D. Oestrogen and progesterone
E. Follicle stimulating hormone (F.S.H) and oestrogen

viii. Green plants are known as producers because they....

A. Produce chlorophyll in chloroplast
B. Produce green leaves
C. Contain fruits and seeds
D. Grow in fertile soil
E. Make food from simple substances ( )

ix. Which of the following prevents surfaces of the articulating bones from being worn out by friction?

A. capsular ligament
B. Synovial membrane
C. Articular Cartilage
D. Tendons
E. Synovial capsule

x. Which of the following is the best definition of allergy?

A. Body reaction to antigen
B. Body reaction against specific drugs
C. Body reaction against a disease-causing micro organism
D. Accumulation of mucus caused by inhaling dust particles
E. Formation of rash caused by eating specific food ( )

2. Match the responses in LIST B with the word or phrases in LIST A by writing the letter of the correct responses beside the item number.

Item no	**LIST A**	**LIST B**
i	Brings about contraction of the uterus at parturition (birth) cause expulsion of milk from mammary glands.	A. Anti-diuretic hormone(A.D.H)
ii	Promote conversion excess glucose in blood into glycogen.	B. Oestrogen
iii	Prepare the body for emergency.	C. Insulin hormones
iv	Stimulate kidneys to absorb water.	D. Glucagon Hormone
v	Have well defined head, thorax and abdomen.	E. Oxytocin hormone
		F. Progesterone hormone
		G. Adrenal cortical hormone
		H. Adrenalin and noradrenalin I. Insecta

SECTION B (60 marks)

3(a) Draw and mention the uses of the following biology laboratory instruments.

(i) Fish net
(ii) Pooter
(iii) Electronic balance
(iv) Thermometer
(v) Test tube rack

3. (b)Explain briefly five rules of biology laboratory.

4. (a) Sir Juma is always suffering from sudden, involuntary and painful contractions of muscles when he attempts strenuous exercise. What is a name of the condition suffered from?

(b) Give three causes of the condition named above (4a)

(c) Give three ways you can advise sir Juma on how to avoid this problem.

5. (a) You as a form four student, you have been studied about types of DNA and RNA, so now provide three differences between them.

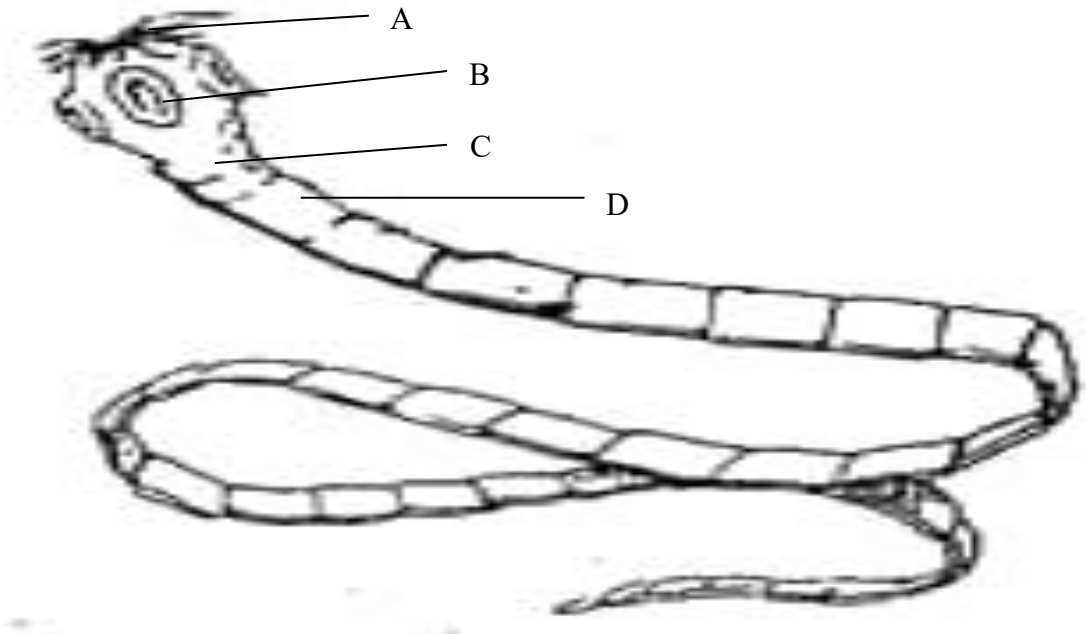
(b) Describe briefly about four composition of DNA.

(c) Give the meaning of the following terms

(i) Phenotype
(ii)Genotype
(iii) Incomplete dominant

6. (a) Explain four points in how insects are beneficial to human beings.

(b) In figure below represent one of the members of the five kingdoms of classification, observe it and then answer the question that follows;



(i) Identify organism x by its common name.
(ii) Label parts A-D

(iii) State the habitat of the specimen x.
(iv) Classify specimen x to class level
(v) State three adaptations of specimen x

7. (a) Define evolution
(b) Explain briefly any four (4) theories of the origin of life.
(c) Outline two (2) Ideas of the Lamark's theory.

8. Mr. Harrison is a botanist whose botanic garden has varieties of flowering plants which produce flowers of different colours. In his attempts to increase the number of Red flowered plants he pollinated a Red Rose plant with a White flowered Rose plant. Unfortunately all f_1 plants produced were Pink flowered plants, which were not in the target of Mr. Harrison.
(i) Use genetic diagram to illustrate the result for f_1 and f_2 generation produced from Mr. Harrison's work.
(ii) Name the pattern of inheritance observed in the cross above.
(iii) Why Pink colour was formed in f_1 generation?

9. (a) State three(3) factor affecting gaseous exchange in animals.
(b) Why tadpole dies when taken out from water to land?

10. (a) Why transportation of materials in living things is important? Give four (4) points
(b) Is water important in living things provide four (4) points

11. Majority of people believe that the study of living things is a wastage of time, as an a biologist clears this misconception. Provide four (4) points.

12. (a) Name the components of human blood.
(b) State the functions of the three (3) blood cells.

SECTION C (25 marks)
Answer two (2) questions form this section. Question number 13 is compulsory

13 (a) Explain four (4) factors affecting body immunity.
(b) Why is necessary to vaccinate people against disease before rather than wait until they get disease? Three (3) points. (15 marks)

14 Write an essay on genetic engineering using the following guidelines
(i) Meaning of genetic engineering
(ii)Advantages or application of genetic engineering
(iii) Disadvantages of genetic engineering (10 marks)

15. Explain six (6) economic importance of kingdom animalia to human being. (10 marks)

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